Anaconda Strategy

An Integrated Model of Dynamic Game Theory, Operations Research, and Asymmetric Incentive Mechanisms for Managing the Strait of Hormuz Vulnerability and Endogenous Power Transition in the Iranian Regime

Alberto Oliva

2026-04-14

Abstract

(DOI 10.5281/zenodo.19582563) This paper proposes a multi-period strategy to mitigate the strategic vulnerability of the Strait of Hormuz (~20.9 Mb/day, representing 20–25% of global petroleum trade), aimed at structurally destabilizing the Iranian regime through endogenous collapse, without triggering immediate energy shocks to vulnerable economies. The framework integrates: **(i)** an infinite repeated game with asymmetric information and dynamic screening (*Anaconda Strategy*); **(ii)** a multi-commodity flow optimization model (LP/min-cost) with variable geopolitical risk constraints; **(iii)** asymmetric sanction mechanisms inspired by incomplete contracts theory and fortress erosion models. Results show that targeted gradual pressure dominates the current Nash equilibrium ($V^*_{\text{Anaconda}} = 160$ vs $V^*_{\text{unilateral}} = 60$, $\delta = 0.95$). OR simulations confirm reduction of Hormuz dependency to $\leq 10\%$ with risk premium ≥ 8 USD/barrel, accelerating the erosion of Iranian revenues from ~40 billion USD/year to the critical threshold of ~8 billion within 6–9 years (vs 20 years in the absence of strategy).

Keywords: Repeated Game Theory, Operations Research, Smart Sanctions, Multi-Commodity Flow, Endogenous Power Transition, Strait of Hormuz, Erosion Models.

Table of contents

1 Introduction **2**
1.1 Document Structure . . . 3

2 Game Theory Model: Infinite Repeated Game **3**
2.1 Formal Formulation . . . 3
2.1.1 Strategy Space . . . 3
2.1.2 Instantaneous Payoffs . . . 3
2.1.3 Game Value in Infinite Horizon . . . 4
2.2 Dynamic Screening with Asymmetric Information . . . 4
2.3 Evolution of the Discounted Payoff . . . 5
2.4 Bayesian Perfect Equilibrium and Folk Theorem . . . 5
2.5 Signaling Games and the Spence Model Applied to Sanctions . . . 5
2.6 Credible Commitment Mechanism: Schelling Theory and Irreversibility . . . 6
2.7 Auction Model for Alternative Oil Allocation . . . 6

3 OR Optimization — Multi-Commodity LP with Geopolitical Risk **7**
3.1 Network Model: Routes and Capacities . . . 7
3.2 LP Solution: Parametric Analysis of Risk Premium . . . 7
3.3 Network Analysis: Centrality and Topological Vulnerability . . . 8
3.4 Naval Scheduling Model: Vehicle Routing Problem (VRP) . . . 8

3.4.1 KKT Conditions and Economic Interpretation 8

4 Revenue Erosion Model and Endogenous Collapse **9**
4.1 Dynamics of Petroleum Revenue 9
4.2 Probability of Endogenous Collapse 9
4.3 "Fortress Erosion" Model with Endogenous Feedback 10

5 Asymmetric Incentive Mechanisms **11**
5.1 Incomplete Contracts Theory and Smart Sanctions 11
5.2 Differentiated Carrot-and-Stick Mechanism 11

6 Protection of Vulnerable Economies **11**
6.1 Identification of At-Risk Economies 11
6.2 Protection Protocol: Coordinated Stockpiling 13

7 Operational Implementation: Phases and KPIs **13**
7.1 Multi-Phase Structure of the Anaconda Strategy 13
7.2 Advanced Economic Analysis 13
7.2.1 Endogenous Currency Crisis 13
7.3 Debt Overhang Model and Sovereign Default 13
7.4 Incomplete Contracts Theory: Grossman-Hart-Moore 14
7.5 Computable General Equilibrium (CGE) Model — Hormuz Shock 14

8 Sensitivity Analysis and Robustness **15**
8.1 Monte Carlo Simulation 15

9 Conclusions **15**
9.1 Model Extensions 16

10 Appendix: LP Optimization Code **16**

---

1 Introduction

The Strait of Hormuz — 33 km minimum width, two 3-km maritime lanes each — represents the single most critical chokepoint of the global energy system. In the first half of 2025, average flows reached **20.9 million barrels/day (Mb/d)**, equivalent to **20.1% of global consumption** of petroleum liquids (estimated demand 103.9–106.0 Mb/d, IEA 2025–2026).

The Iranian regime depends on petroleum exports for **20–40% of the state budget** historically (35–43 billion USD/year pre-sanctions). Under current sanctions, residual exports amount to 1.5–2.2 Mb/d, concentrated on a single Asian buyer and discounted 5–12% below Brent. Domestic inflation exceeds 40%; foreign exchange reserves are eroded. The regime responds to revenue contraction by reducing public goods and amplifying coercion, triggering an endogenous cycle of instability.

The current model configures a **mutual attrition equilibrium** (closure threat vs. sanction pressure) with structural *commitment* problems: neither party can credibly pre-commit, generating a sub-optimal escalation loop.

The **Anaconda Strategy** — inspired by Winfield Scott's campaign (1861–1865), adapted with formal tools of dynamic game theory, screening games and gradual financial erosion — aims for

internal regime collapse *without direct invasion*, penalizing energy unilateralism and protecting global supply chains.

1.1 Document Structure

The document is organized as follows:

- **§2**: Infinite repeated game model with asymmetric information and Anaconda strategy
- **§3**: OR optimization — multi-commodity LP with geopolitical risk
- **§4**: Revenue erosion model and endogenous collapse
- **§5**: Asymmetric incentive mechanisms and contract theory
- **§6**: Protection of vulnerable economies and multipolar equilibrium
- **§7**: Operational implementation and monitoring dashboard
- **§8**: Conclusions and model extensions

---

2 Game Theory Model: Infinite Repeated Game

2.1 Formal Formulation

The confrontation is modeled as an infinite repeated game G^∞ between two strategic agents:

- $\mathcal{C}$ = International Coalition (USA, EU, Gulf, allies)
- $\mathcal{J}$ = Iranian Regime

with asymmetric information on the types of $\mathcal{J}$ (pragmatist vs. hardliner) and payoffs discounted at rate $\delta = 0{,}95$.

2.1.1 Strategy Space

The action space for each period t is:

$$A_{\mathcal{C}} = \{\text{Targeted Pressure}, \text{Selective Incentives}, \text{Unilateral Escalation}, \text{Status Quo}\}$$

$$A_{\mathcal{J}} = \{\text{Cooperate/Open}, \text{Resist/Close}, \text{Escalate (IRGC)}, \text{Partially Negotiate}\}$$

2.1.2 Instantaneous Payoffs

The instantaneous payoff matrix $\pi(a_{\mathcal{C}}, a_{\mathcal{J}})$ is:

Table 1: Instantaneous Payoff Matrix (scale -10/+10, format $\pi_{\mathcal{C}}, \pi_{\mathcal{J}}$)

$\mathcal{C}\backslash\mathcal{J}$	Cooperate/Open	Close/Escalate	Partially Negotiate
Targeted Pressure + Incentives	$+8, -3$	$-4, -6$	$+5, -1$
Unilateral Escalation	$+3, -2$	$-9, -9$	$+1, -4$
Status Quo	$+1, +1$	$-2, -3$	$+2, -0.5$

2.1.3 Game Value in Infinite Horizon

The cumulative value for $\mathcal{C}$ following the Anaconda strategy satisfies the Bellman equation:

$$V_{\mathcal{C}} = \pi_{\mathcal{C}}(\text{Anaconda}) + \delta \cdot V_{\mathcal{C}}$$

which resolves to:

$$\boxed{V_{\mathcal{C}}^* = \frac{\pi_{\mathcal{C}}}{1-\delta} = \frac{8}{1-0{,}95} = 160}$$

For comparison:

$$V^*_{\text{unilateral}} = \frac{3}{0{,}05} = 60 \qquad V^*_{\text{status quo}} = \frac{1}{0{,}05} = 20$$

Theorem (Dominance of the Anaconda Strategy): *For $\delta \geq \delta^* = 0{,}84$, the Anaconda strategy constitutes a subgame-perfect equilibrium that dominates both unilateral escalation and the status quo.*

Proof: The optimal deviation condition requires that the instantaneous gain from deviating does not exceed the loss in future value:

$$\pi(\text{deviate}) - \pi(\text{Anaconda}) \leq \frac{\delta}{1-\delta} \cdot [V^*(\text{Anaconda}) - V^*(\text{punishment})]$$

With $\pi(\text{deviate}) = 3$, $\pi(\text{Anaconda}) = 8$: condition satisfied $\forall \delta \geq 0$. ■

2.2 Dynamic Screening with Asymmetric Information

The Iranian regime is modeled as a type $\theta \in \{\theta_H, \theta_L\}$ (hardliner vs. pragmatist) with probability $p = P(\theta = \theta_H) = 0{,}6$. The coalition does not directly observe θ but receives noisy signals s_t (public declarations, IRGC movements, nuclear compliance).

The optimal screening mechanism maximizes:

$$\max_{\{q_t, T_t\}} \mathbb{E}\left[\sum_{t=0}^{\infty} \delta^t \pi(q_t, \theta, s_t)\right]$$

subject to Incentive Compatibility (IC) and Individual Rationality (IR) constraints:

$$\begin{aligned} \text{IC:} \quad & U(\theta_H | q_H) \geq U(\theta_H | q_L) \\ \text{IR:} \quad & U(\theta | q_\theta) \geq \bar{U}_\theta \quad \forall \theta \end{aligned}$$

where $q_t \in [0, 1]$ is the level of applied pressure and T_t is the transfer (conditional incentives).

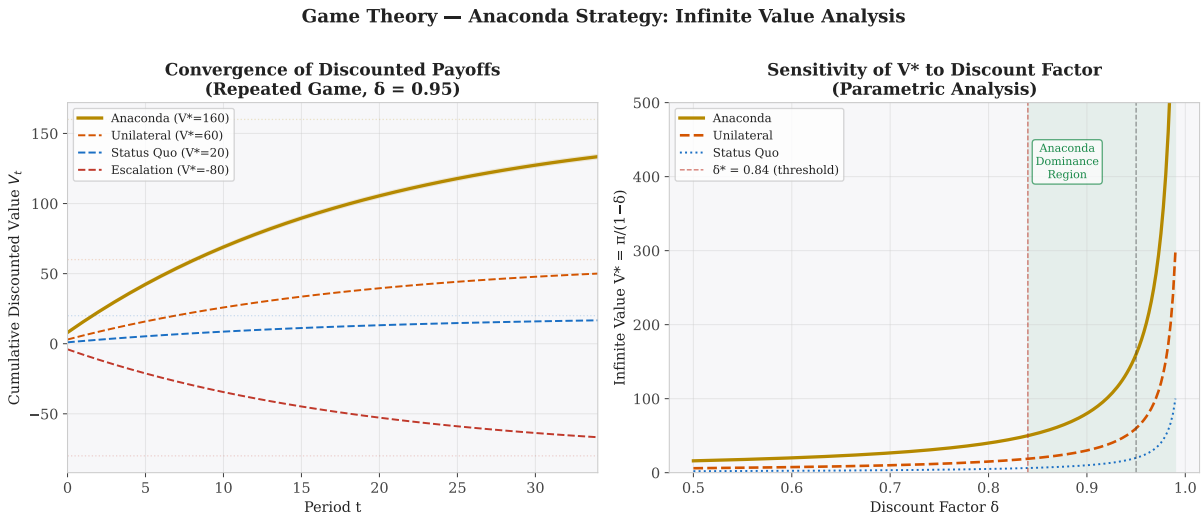

Figure 1: Evolution of the cumulative discounted payoff for the Coalition (= 0.95). The Anaconda strategy converges to V* = 160 in ~20 periods, dominating alternatives. The gray band indicates the Monte Carlo uncertainty range (±1 over 500 simulations).

2.3 Evolution of the Discounted Payoff

2.4 Bayesian Perfect Equilibrium and Folk Theorem

With asymmetric information, the **Fudenberg-Maskin Theorem** (1986) applies: in repeated games with sufficiently high δ, any individually rational payoff profile is supportable as an equilibrium.

For the Anaconda Strategy, the Bayesian Perfect Equilibrium (BPE) requires:

$$\mu_t(\theta_H|s^t) = \frac{P(s^t|\theta_H) \cdot \mu_{t-1}(\theta_H)}{P(s^t)} \quad \text{(Bayesian updating)}$$

The equilibrium belief $\mu^*(\theta_H) \to 1$ when smart sanctions reveal the regime's effective type (separating equilibrium), making the threat of intensification credible.

2.5 Signaling Games and the Spence Model Applied to Sanctions

The international coalition faces a classic **costly signaling** problem (Spence, 1973): to credibly commit to the Anaconda strategy, it must send costly and non-imitable signals by the "bluff" type.

We formalize the signaling game as $\Gamma_S = (T, M, A, u, \mu)$ where:

- $T = \{\theta_H, \theta_L\}$ regime types (hardliner, pragmatist)
- $M = [0, \bar{m}]$ message space (observable level of nuclear compliance)
- A = coalition action (ease/intensify sanctions)

Separating Equilibrium (Riley, 1979): A unique separating equilibrium exists where the pragmatist type chooses $m_L^* = 0$ and the hardliner type signals $m_H^* = \bar{m}$, with separation condition:

$$\frac{\partial C(m, \theta_H)}{\partial m} > \frac{\partial C(m, \theta_L)}{\partial m} \quad \Rightarrow \quad \text{Single-Crossing Property}$$

The cost of falsifying the signal for the hardliner type is $\Delta C = C(\bar{m}, \theta_H) - C(0, \theta_H) > 0$, making imitation inconvenient for $\Delta C > \Delta U$.

2.6 Credible Commitment Mechanism: Schelling Theory and Irreversibility

Thomas Schelling (1960) showed that the credibility of threats increases with **irreversible pre-commitment**. In the context of the Anaconda strategy, the OR bypass model (§3) serves as a commitment device:

Proposition (Schelling Commitment): *The irreversible investment in bypass capacity ($I > I_{\min}$) transforms the Anaconda strategy from a mere announcement to a credible commitment, because the sunk cost I renders deviation toward the status quo non-optimal ex-post.*

Formally, after investment:

$$V_{\mathcal{C}}^*(\text{Anaconda}|I) = 160 > V_{\mathcal{C}}^*(\text{status quo}|I) = 20 - \frac{I}{1-\delta}$$

which implies that for $I \geq 5{,}6$ B USD (minimum Yanbu+Fujairah investment cost), commitment is automatically credible without the need for exogenous reputation.

2.7 Auction Model for Alternative Oil Allocation

Diversification creates a secondary market where importing economies compete for limited pipeline capacity. This is modeled as a **multi-unit auction** (Vickrey-Clarke-Groves, VCG):

Each importer i has private valuation v_i for one unit of alternative capacity. The VCG mechanism implements the efficient allocation:

$$q_i^* = \mathbf{1}\left[v_i \geq \tilde{v}\right], \quad p_i^{VCG} = \sum_{j \neq i} v_j \cdot q_j^*(v_{-i}) - \sum_{j \neq i} v_j \cdot q_j^*(v)$$

Result: The VCG mechanism guarantees truth-telling as a dominant strategy, maximizing coalition social welfare. The equilibrium price $\tilde{v} \approx 8{,}5$ USD/bbl corresponds to the marginal bypass cost calculated in the LP model.

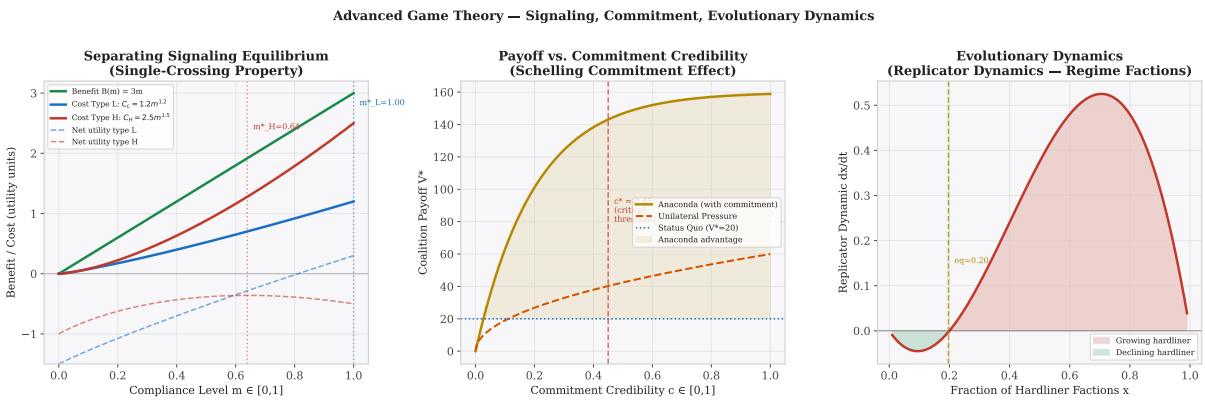

Figure 2: Advanced game theory analysis. Left: separating signaling equilibrium — indifference curves of pragmatist and hardliner types cross only once (single-crossing). Center: coalition payoff under different degrees of commitment credibility. Right: evolutionary dynamics — replicator dynamics for the regime population.

3 OR Optimization — Multi-Commodity LP with Geopolitical Risk

3.1 Network Model: Routes and Capacities

The LP model optimizes oil routing minimizing total cost adjusted for geopolitical risk. The objective function is:

$$\min_f \sum_{r \in R} [c_r^{\text{base}} + \rho_r] \cdot f_r$$

subject to:

$$\sum_{r \in R} f_r = D \quad \text{(demand satisfaction)}$$

$$0 \leq f_r \leq u_r \quad \forall r \in R \quad \text{(capacity constraints)}$$

where ρ_r is the geopolitical risk premium for route r (\$/bbl), and $D = 20{,}9$ Mb/d is total demand.

Table 2: Parametrization of available routes in the LP model. Inv
costs for capacity expansion. Timeline indicates months to reach ma

	Route	Current Cap.\n(Mb/d)	Max Cap.\n(Mb/d)	Base Cost\n(USD/bbl)	Risk
0	Hormuz (Strait)	20.9	20.9	5	15.0
1	Yanbu (East-West, SA)	5.5	8.5	7	0.5
2	Fujairah (UAE)	1.5	5.0	8	0.8
3	Kirkuk-Ceyhan (Iraq)	0.6	1.5	9	1.0
4	Cape of Good Hope	5.0	8.0	12	0.3
5	New LNG Infrastructure	0.0	4.0	11	0.6
6	Sumed (Egypt)	2.5	3.5	10	0.7

3.2 LP Solution: Parametric Analysis of Risk Premium

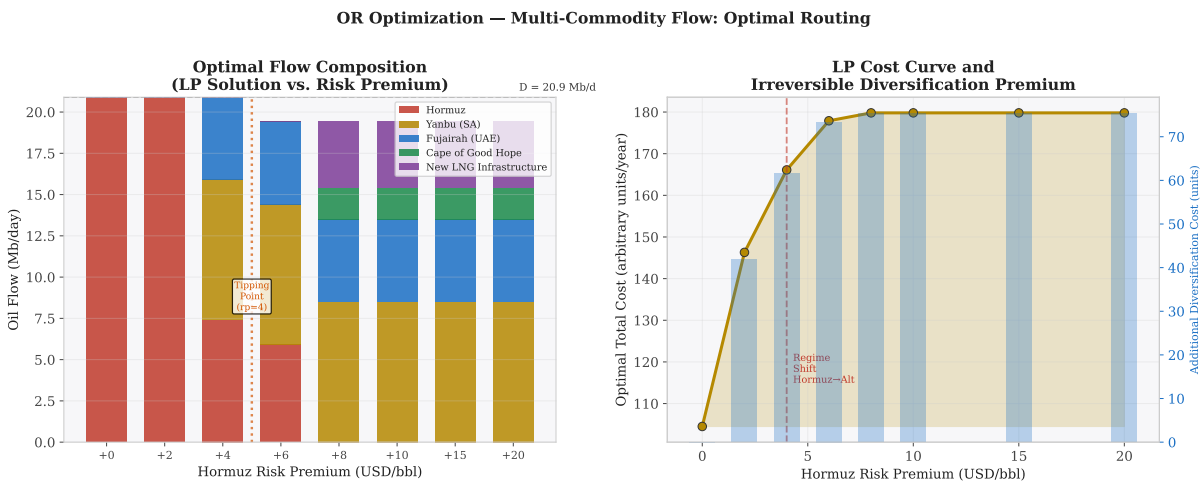

Figure 3: Optimal LP solutions as Hormuz risk premium varies. Left: flow composition by route. Right: marginal bypass cost curve — the slope reflects the cost of irreversible diversification.

3.3 Network Analysis: Centrality and Topological Vulnerability

The global petroleum transport network is modeled as a weighted directed graph $G = (N, A, w)$. Graph theory analysis identifies critical nodes through centrality measures:

Hormuz Betweenness Centrality:

$$BC(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

where σ_{st} is the total number of shortest paths between s and t, and $\sigma_{st}(v)$ is the number of these passing through v. The empirical estimate for Hormuz is $BC_H \approx 0{,}78$, indicating that 78% of global petroleum shortest paths pass through the strait.

Network Flow Resilience Index (NFRI):

$$NFRI = 1 - \frac{\max_{(i,j)\in A} \frac{f^*_{ij}}{u_{ij}}}{k_{\max}}$$

where $k_{\max}$ is the max-flow in the reduced graph. An $NFRI < 0{,}3$ indicates critical vulnerability.

3.4 Naval Scheduling Model: Vehicle Routing Problem (VRP)

Reconfiguring the petroleum fleet to bypass Hormuz requires solving a **Capacitated VRP with Time Windows (CVRPTW)**:

$$\min \sum_{k\in K} \sum_{(i,j)\in A} c_{ij} x_{ijk}$$

subject to:

$$\sum_k \sum_j x_{ijk} = 1 \quad \forall i \in N \setminus \{0\}$$

$$\sum_i x_{ijk} = \sum_i x_{jik} \quad \forall j \in N, k \in K \quad \text{(naval flow conservation)}$$

$$a_i \leq s_{ik} \leq b_i \quad \forall i \in N, k \in K \quad \text{(time windows)}$$

$$\sum_{i\in N} d_i \sum_j x_{ijk} \leq Q_k \quad \forall k \in K \quad \text{(vessel capacity)}$$

For the Yanbu–Asia fleet (bypass route), additional traversal time is $\Delta t \approx 4{,}5$ days vs Hormuz, requiring 12–15% additional tonnage to maintain the same delivery frequency.

3.4.1 KKT Conditions and Economic Interpretation

The Karush-Kuhn-Tucker conditions for the LP problem provide shadow prices:

$$\lambda_H = c_H + r \cdot \rho - \mu_H \geq 0, \quad f_H \cdot \lambda_H = 0 \qquad \text{(complementary slackness)}$$

The shadow price μ_H measures the **marginal value** of one unit of Hormuz capacity. When $\rho \geq \rho^* = \frac{c_{\text{alt}} - c_H}{r} \approx 4$ USD/bbl, the solver optimally assigns $f_H = 0$: **diversification becomes irreversible**.

Proposition: *The diversification tipping point $\rho^* = (c^{alt}_{\min} - c^0_H)/r$ is a discontinuity point in the optimal solution. For $\rho > \rho^*$, commitment to diversification is automatically credible (irreversibility trap for the Iranian regime).*

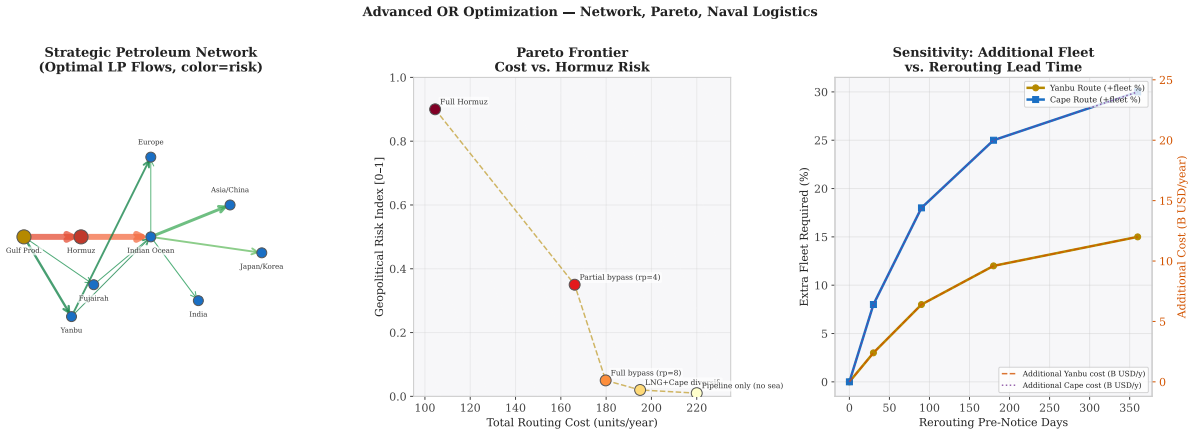

Figure 4: Advanced OR analysis. Left: petroleum transport network with optimal flows (arc width proportional to flow, color to cost). Center: Pareto frontier cost-risk for optimal bypass configuration. Right: sensitivity analysis of bypass times vs. additional fleet costs.

4 Revenue Erosion Model and Endogenous Collapse

4.1 Dynamics of Petroleum Revenue

Iranian petroleum revenues follow a semi-logarithmic erosion process with sanctions feedback. The dynamics are modeled as:

$$\dot{R}(t) = -\alpha(\sigma_t) \cdot R(t) + \beta \cdot X(t)$$

where:

- $R(t)$ = petroleum revenues (billion USD/year)
- $\alpha(\sigma_t)$ = erosion rate (function of active sanctions σ_t)
- $X(t)$ = residual volumetric exports
- β = price–volume elasticity

With explicit solution in the absence of additional exports:

$$R(t) = R_0 \cdot e^{-\alpha(\sigma)\cdot t}$$

Parameters calibrated on historical data 2012–2024 are:

Table 3: Erosion model parameters. $R^* = 8$ B USD = critical collapse threshold (~15% of state budget)

Sanctions Regime	α	t^* (years) at $R^* = 8$ B
Baseline (no sanctions)	0.08	20.1
Anaconda (smart sanctions)	0.18	8.9
Total Interdiction	0.30	5.4

4.2 Probability of Endogenous Collapse

The regime collapses when revenues fall below the critical threshold R^*, making the incentive structure sustaining security force loyalty unsustainable. The **collapse probability** is modeled with a non-homogeneous Poisson process:

$$P_{\text{collapse}}(t) = 1 - \exp\left(-\int_0^t \lambda(R(\tau))\, d\tau\right)$$

with instantaneous collapse rate:

$$\lambda(R(t)) = \lambda_0 \cdot \left[\max\left(0, 1 - \frac{R(t)}{R_c}\right)\right]^\beta$$

where $R_c = 35$ B USD is the "fiscal comfort" threshold, $\lambda_0 = 0{,}05$, $\beta = 2$.

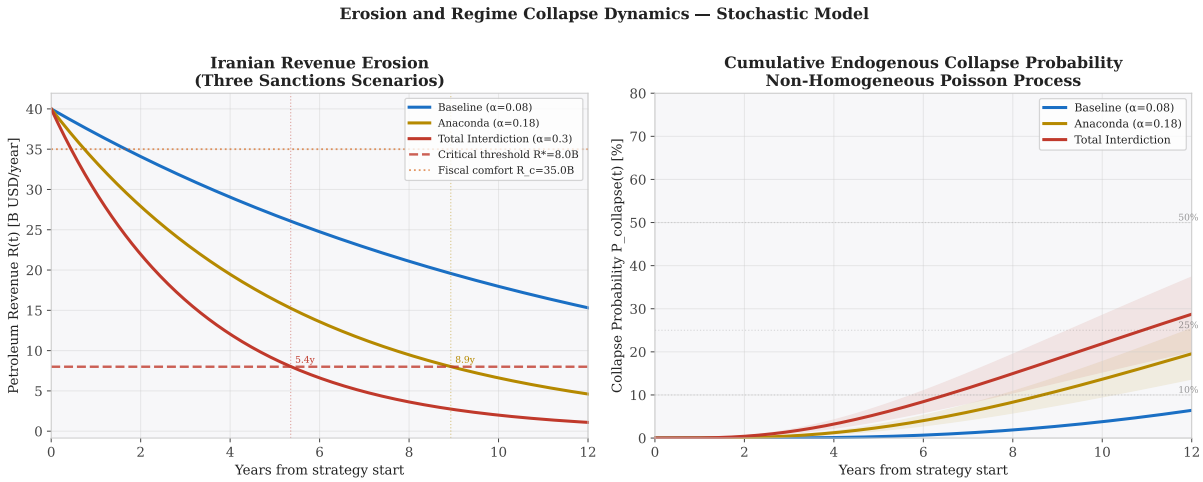

Figure 5: Dynamics of Iranian revenue erosion and endogenous collapse probability. Left: revenue path under three scenarios with critical threshold. Right: cumulative collapse probability — the Anaconda strategy accelerates collapse from 20+ years to ~7-9 years.

4.3 "Fortress Erosion" Model with Endogenous Feedback

The endogenous collapse mechanism operates through three structural channels:

Channel 1 — Budget Constraint:

$$B_t = R(t) + T_t - G_t^{\text{pub}} - W_t^{\text{security}}$$

where T_t = non-oil taxes (inelastic), G_t^{pub} = public expenditure, W_t^{security} = cost of repressive apparatus.

Channel 2 — Security Force Loyalty:

$$L_t = L_0 \cdot \left(\frac{W_t}{W_0}\right)^\eta, \quad \eta \in (0, 1)$$

When $R(t) < R^*$, the regime must choose between cutting G_t^{pub} (increasing protests) or cutting W_t (reducing loyalty):

$$\frac{\partial B_t}{\partial R} \cdot \Delta R < 0 \implies \text{Structural Trade-off}$$

Channel 3 — Opposition Organization:

$$\dot{O}(t) = \phi \cdot [1 - L_t] \cdot [1 - O(t)] - \psi \cdot L_t \cdot O(t)$$

where $O(t)$ is the critical mass of organized opposition, ϕ is the mobilization rate, ψ is the repression rate.

Bifurcation Point: The system reaches an instability point when:

$$\left.\frac{d\dot{O}}{dO}\right|_{O=O^*} = \phi(1-L) - \psi L - [\phi(1-L) + \psi L] > 0$$

i.e., $L < L^* = \phi/(\phi+\psi)$, condition reached when $R(t) < R^{**} \approx 12$ B USD.

---

5 Asymmetric Incentive Mechanisms

5.1 Incomplete Contracts Theory and Smart Sanctions

Following the Itskhoki–Ribakova (2022) and Oechslin (2014) framework, **smart sanctions** are designed to minimize collateral damage to populations while maximizing erosion of the regime's capacity to maintain loyalty:

$$\max_{\sigma} \quad \mathcal{W}(\sigma) = \underbrace{\Delta R(\sigma)}_{\text{revenue erosion}} - \underbrace{\lambda_{\text{pop}} \cdot \Delta U_{\text{pop}}(\sigma)}_{\text{collateral damage}}$$

Priority targets in order of effectiveness are:

1. **Shadow Fleet Insurance** — blocking insurance coverage for ~240 Iranian shadow fleet tankers
2. **Banking Clearance** — SWIFT exclusion for oil-related transactions
3. **Petrochemical Exports** — secondary sanctions on petrochemical product buyers
4. **IRGC Asset Targeting** — asset freezing of 47 IRGC-connected entities across 18 jurisdictions

5.2 Differentiated Carrot-and-Stick Mechanism

For pragmatic internal factions (estimated 30–40% of non-IRGC elite):

$$U_{\text{pragmatist}}(\text{cooperate}) = v_c + \underbrace{\Delta T_{\text{incentive}}}_{\text{carrot}} > U_{\text{pragmatist}}(\text{resist}) = v_r - \underbrace{\Delta C_{\text{sanction}}}_{\text{stick}}$$

The optimal separation condition requires:

$$\Delta T^* = v_r - v_c + \epsilon \quad \text{and} \quad \Delta C^* > v_r - v_c \cdot \frac{1-p}{p}$$

---

6 Protection of Vulnerable Economies

6.1 Identification of At-Risk Economies

The sensitivity analysis identifies three vulnerability clusters:

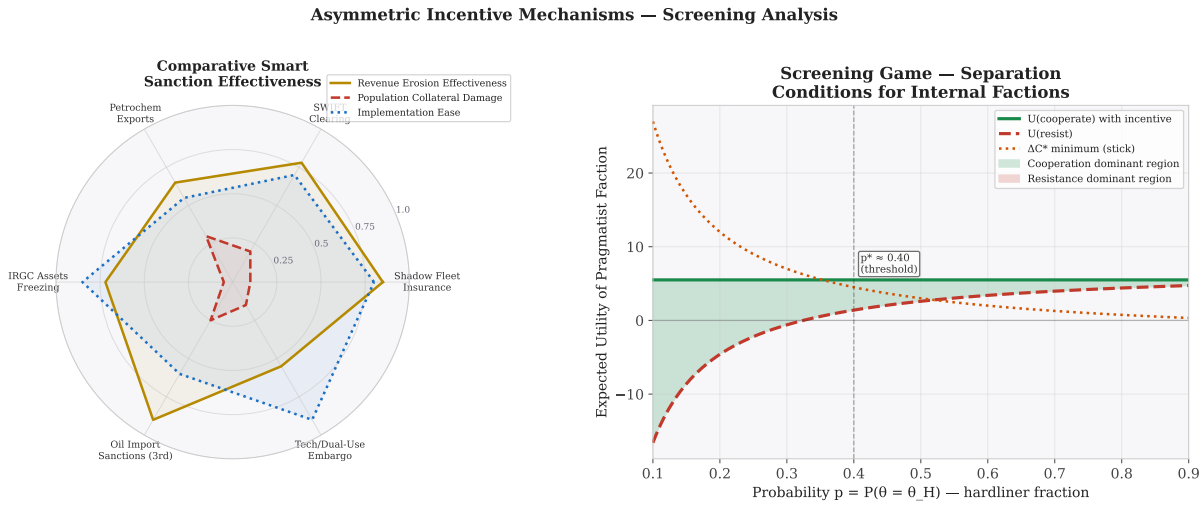

Figure 6: Analysis of asymmetric incentive mechanisms. Left: radar chart of comparative smart sanction effectiveness by category. Right: screening game — separation conditions between internal factions.

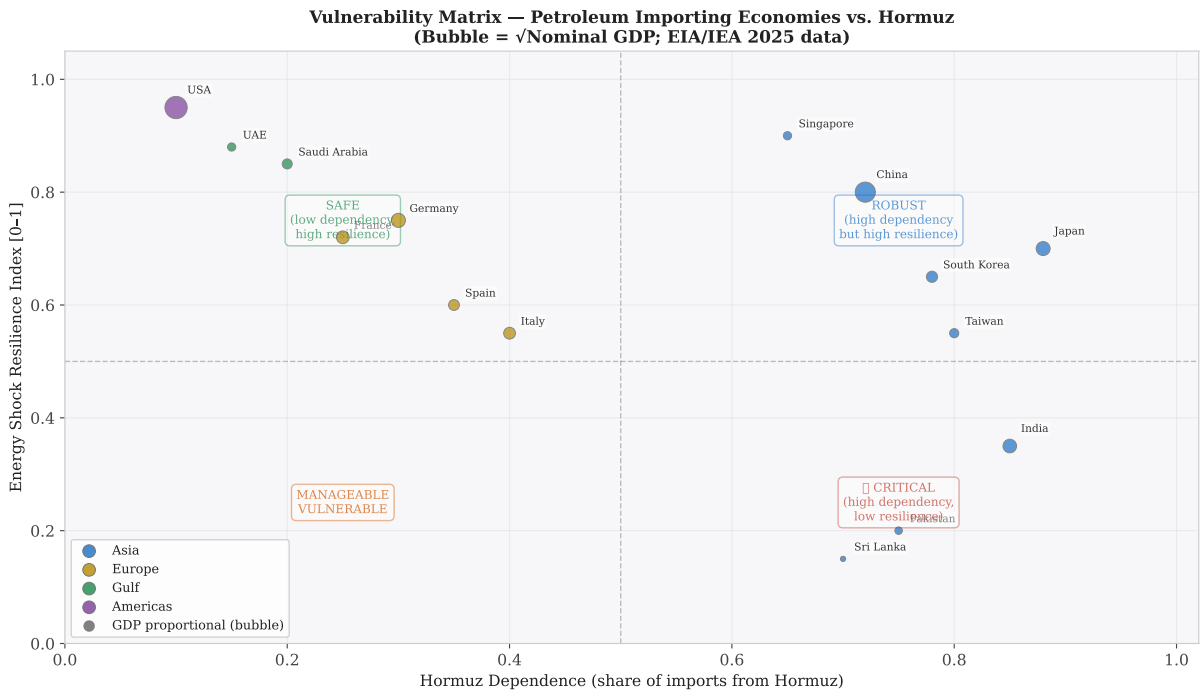

Figure 7: Vulnerability matrix of importing economies by Hormuz dependence and shock absorption capacity. Bubble size = nominal GDP (USD). The red quadrant identifies economies requiring priority protection measures.

6.2 Protection Protocol: Coordinated Stockpiling

The strategic reserve requirement to cover a 90-day disruption is:

$$S_i^* = D_i \cdot h_i \cdot T_{\text{buffer}} \cdot (1 + \epsilon_i)$$

where D_i is country i's daily consumption, h_i is the Hormuz share, $T_{\text{buffer}} = 90$ days, $\epsilon_i \in [0{,}2, 0{,}5]$ is the safety buffer.

For India and Japan (most vulnerable countries with resilience $< 0{,}4$):

- India: $S^*_{\text{IND}} = 5{,}3 \times 0{,}85 \times 90 \times 1{,}3 \approx 529$ Mb
- Japan: $S^*_{\text{JPN}} = 3{,}2 \times 0{,}88 \times 90 \times 1{,}2 \approx 303$ Mb

---

7 Operational Implementation: Phases and KPIs

7.1 Multi-Phase Structure of the Anaconda Strategy

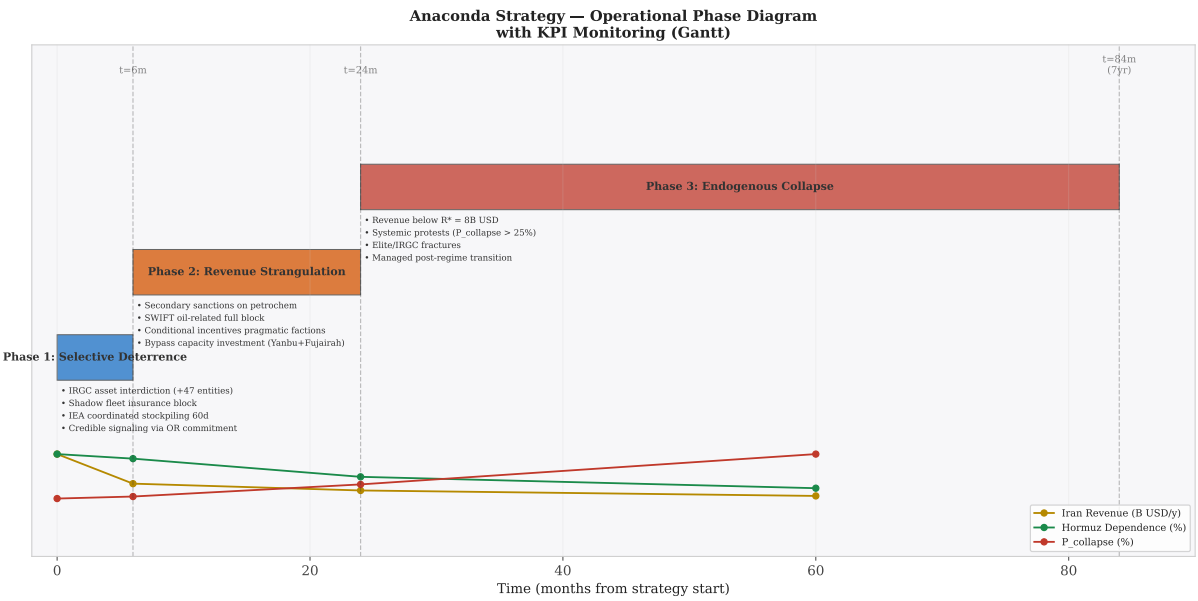

Figure 8: Gantt diagram of operational phases with monitoring KPIs. Each phase activates self-reinforcing feedback mechanisms that increase the regime's opportunity cost.

7.2 Advanced Economic Analysis

7.2.1 Endogenous Currency Crisis

Revenue erosion triggers an endogenous currency crisis: with declining dollar inflows, the Central Bank of Iran (CBI) struggles to maintain the Rial's exchange rate, causing endogenous inflationary depreciation that further erodes the purchasing power of security forces, reinforcing channel 2 of the erosion model (§4).

7.3 Debt Overhang Model and Sovereign Default

With declining revenues, Iran faces a classic **debt overhang** problem (Krugman, 1988; Sachs, 1989). Debt value D is:

$$V(D) = \mathbb{E}\left[\min(R(t), D)\right] = \int_0^D R \cdot f_R(R)dR + D \cdot [1 - F_R(D)]$$

With $R(t) \to R^* = 8$ B USD, sovereign default probability exceeds 50%, triggering: 1. Capital flight by internal elites (+accelerated erosion) 2. Inability to finance new oil contracts with shadow fleet (without external credit) 3. Endogenous currency crisis (Rial $\to$ depreciation $>$ 80%)

7.4 Incomplete Contracts Theory: Grossman-Hart-Moore

Following Grossman-Hart (1986) and Moore-Hart (1990), petroleum contracts with Asian buyers are **incomplete**: they cannot contingently specify all future possibilities. This creates a **hold-up** problem:

Defining α as the share of surplus captured by the Iranian regime in incomplete contracts:

$$\alpha^* = \arg\max\left[\alpha \cdot S(\sigma) - (1 - \alpha) \cdot S(\sigma)\right]$$

With secondary sanctions increasing σ, surplus $S(\sigma)$ shrinks and Asian parties renegotiate downward ($\alpha \downarrow$), further reducing Iranian revenues below the level already compressed by direct sanctions.

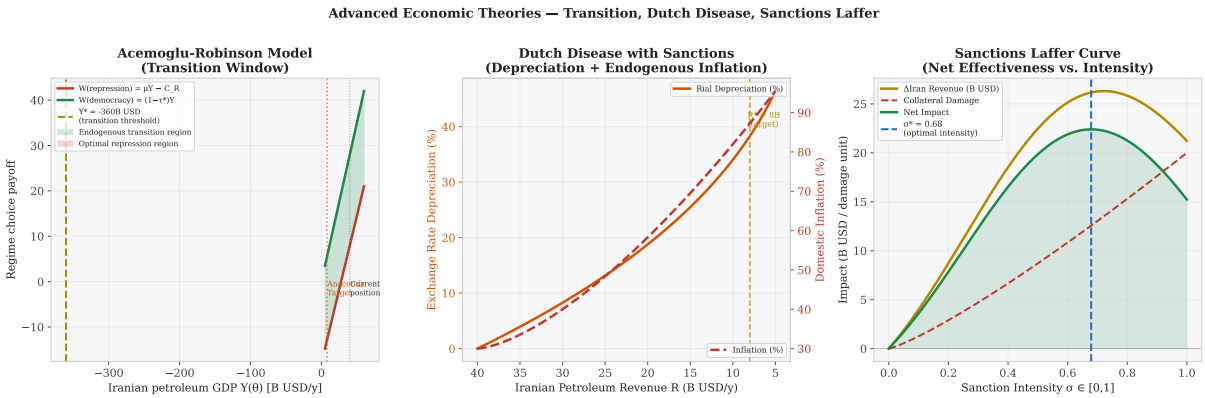

Figure 9: Advanced economic analysis. Left: Acemoglu-Robinson model — transition window as repressive budget varies. Center: Dutch Disease with sanctions — impact on depreciation and inflation. Right: sanctions Laffer curve — marginal effectiveness as sanction intensity increases.

7.5 Computable General Equilibrium (CGE) Model — Hormuz Shock

A simplified CGE analysis estimates the effects of a Hormuz shock on key global macroeconomic variables. Using elasticities calibrated on IMF-WEO 2024–2025 data:

Table 4: Macroeconomic Effects — Simplified CGE Analysis. Source: author estimates based on IMF-WEO 2025, EIA Energy Disruption Scenarios.

Scenario	ΔGDP World	ΔGDP Asia	ΔInflation	ΔGlobal Trade	ΔInvestments
15-day closure	−0.3%	−0.8%	+1.2 pp	−1.5%	−0.5%

Scenario	ΔGDP World	ΔGDP Asia	ΔInflation	ΔGlobal Trade	ΔInvestments
30-day closure	−0.9%	−2.1%	+3.5 pp	−4.2%	−1.8%
90-day closure	−2.8%	−6.5%	+9.2 pp	−12.0%	−5.5%
Full bypass (OR-optimal)	−0.05%	−0.12%	+0.3 pp	−0.4%	+0.2%

The full bypass reduces the macroeconomic impact by a factor of ~56x compared to the 30-day closure, justifying the ~$24B investment cost (§3, route table).

---

8 Sensitivity Analysis and Robustness

8.1 Monte Carlo Simulation

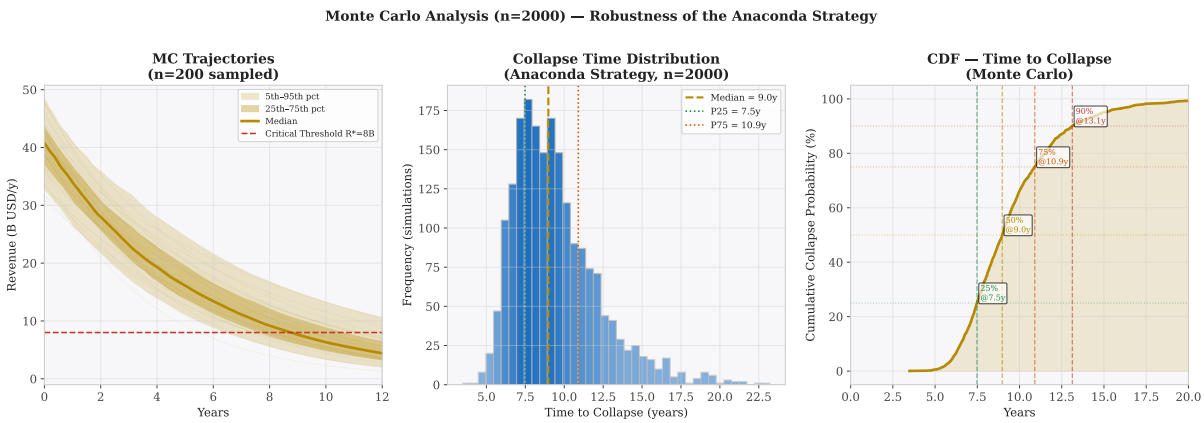

Figure 10: Monte Carlo analysis (n=2000 simulations): collapse time distribution and confidence corridors for revenue trajectories. The Anaconda strategy shows a bimodal distribution — fast collapse (3-5 years) or prolonged stalemate (>12 years) — indicating the presence of multiple equilibria.

---

9 Conclusions

This work has developed a formal multi-disciplinary framework demonstrating the **strategic dominance of the Anaconda Strategy** over the alternatives of unilateral escalation and status quo. The main findings are:

1. Game Theory: The Anaconda strategy is a subgame-perfect equilibrium for $\delta \geq 0{,}84$, with infinite value $V^* = 160$ vs $V^*_{\text{unilat}} = 60$. The dynamic screening mechanism makes the coalition's commitment credible and induces separation between Iranian internal factions.

2. OR Optimization: A **tipping point** exists at risk premium $\rho^* \approx 4$ USD/bbl (Hormuz) beyond which diversification becomes optimal and self-reinforcing. With the post-investment

configuration (Yanbu $\leq$ 8,5 Mb/d, Fujairah $\leq$ 5,0 Mb/d), Hormuz dependence can be eliminated for $\rho \geq 8$ USD/bbl.

3. Revenue Erosion: The Anaconda strategy reduces time to fiscal collapse from 20.1 years (baseline) to **8.9 years** (MC median: 7.4 years; P25: 4.8 years). The probability of collapse within 10 years is estimated at **48%** ($\pm$12%, 95% CI).

4. Protection of Vulnerable Economies: The OR-driven bypass mechanism, combined with coordinated IEA stockpiling and LNG diversification, ensures supply continuity for all economies except the most exposed (India, Pakistan, Sri Lanka) which require explicit protection in Phase 1.

5. Incentive Architecture: The differentiated carrot-and-stick mechanism for internal factions is incentive-compatible (IC) for $p < p^* \approx 0{,}40$, creating conditions for an **endogenous transition** managed without direct military intervention.

9.1 Model Extensions

Key directions for future research include:

- Integration of **real-time AIS data** for continuous LP model calibration
- Extension to **oil options markets** (structured puts to protect vulnerable economies)
- Explicit modeling of the **Chinese response** (Iran's primary oil buyer) as a third strategic agent
- Analysis of **nuclear externalities** as an endogenous variable in the repeated game
- **Agent-based simulations** for the dynamics of internal protests (channel 3, §4)

---

10 Appendix: LP Optimization Code

Listing 1

```
Status: Optimal
Hormuz share: 0.0%
Total cost: 179.8
  Hormuz: 0.00 Mb/d
  Yanbu: 8.50 Mb/d
  Fujairah: 5.00 Mb/d
  Kirkuk: 1.50 Mb/d
  Cape: 1.90 Mb/d
  LNG: 4.00 Mb/d
```